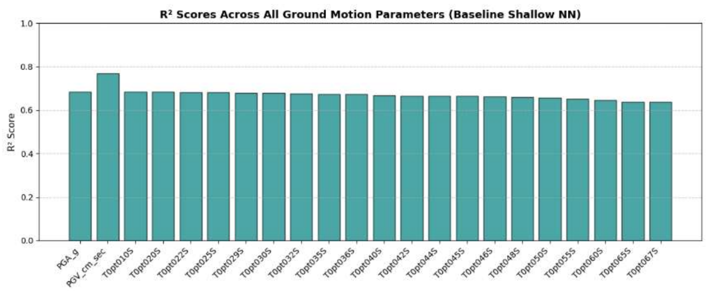
CE 6018 Seismic Data Analysis, Tutorial 4, 28/02/2026

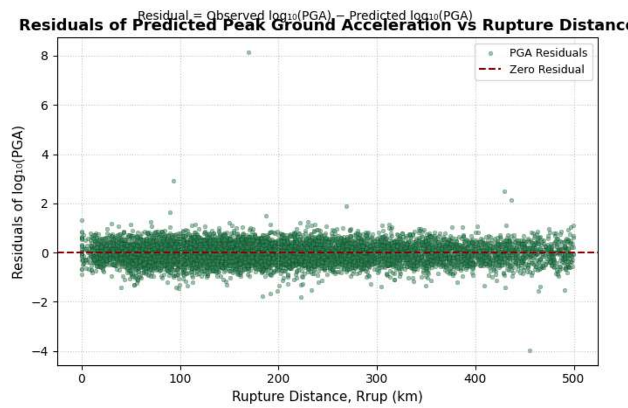
Chintha Vishnu Vardhan CE23B005

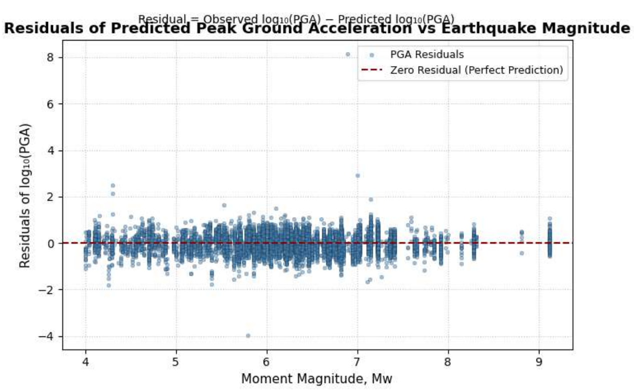
1) Shallow Neural Network

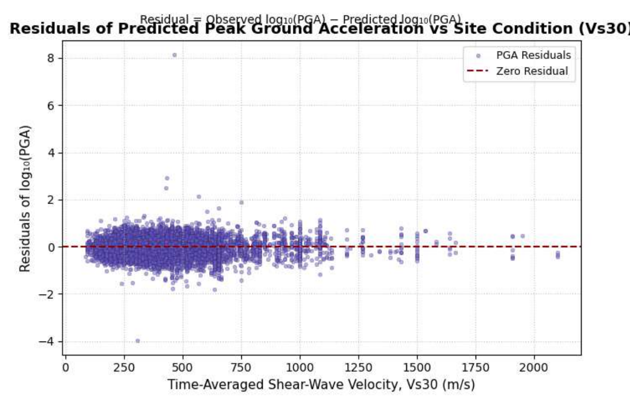
A. Baseline Model

```
model = Sequential([
  Dense(16, activation='relu', input_shape=(X_train.shape[1],)), # Single hidden layer = Shallow NN
  Dense(22, activation='linear')
])
model.compile(optimizer=Adam(learning_rate=0.001), loss='mse', metrics=['mae'])
```

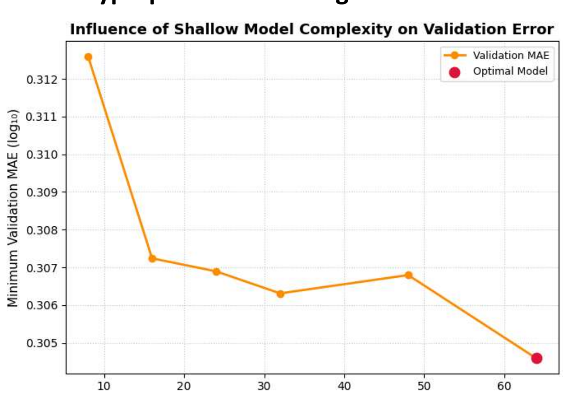


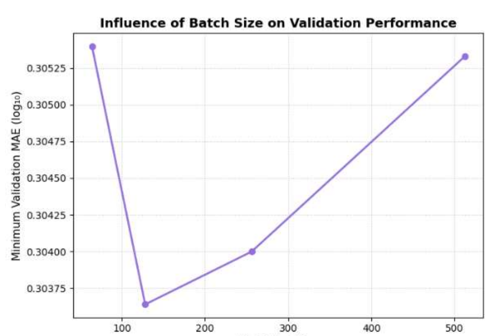


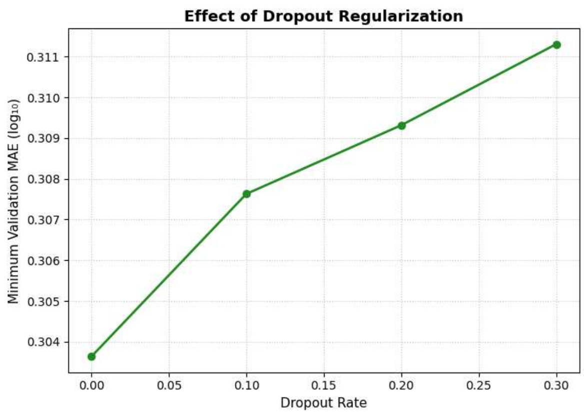


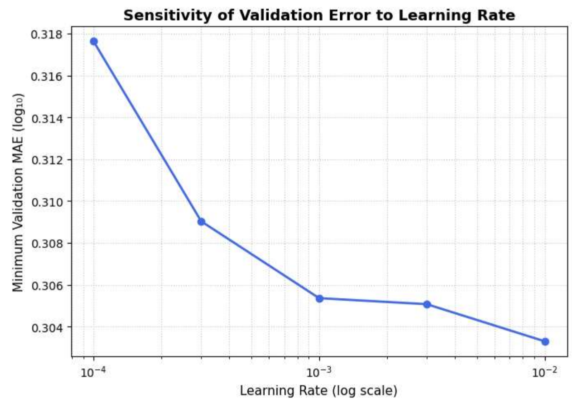


B. Hyperparameter tuning:



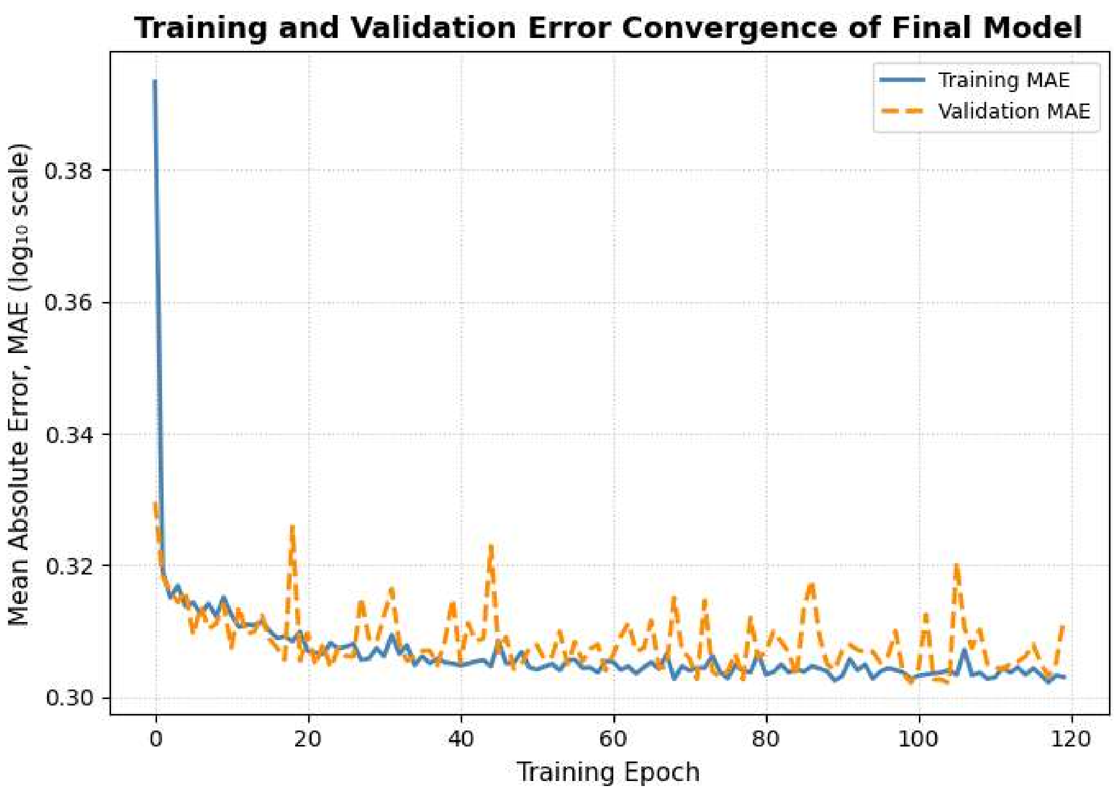






OPTIMAL HYPERPARAMETERS FOUND:
Neurons: 64 | LR: 0.01 | Batch Size: 128 | Dropout: 0.0

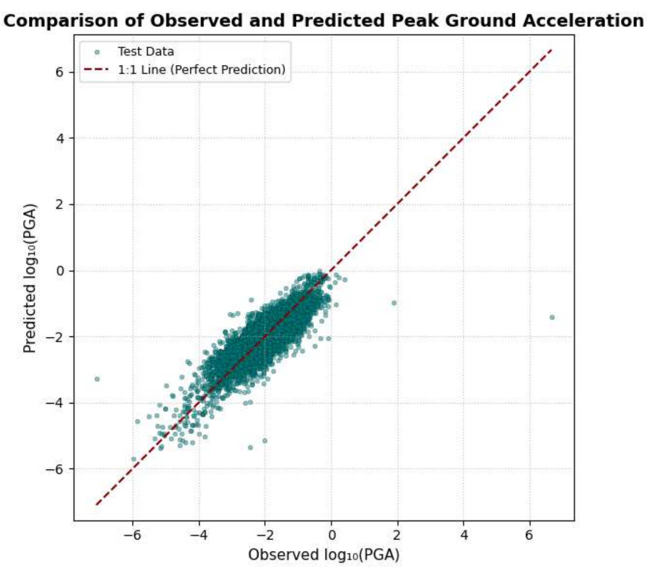
C. FINAL OPTIMIZED MODEL

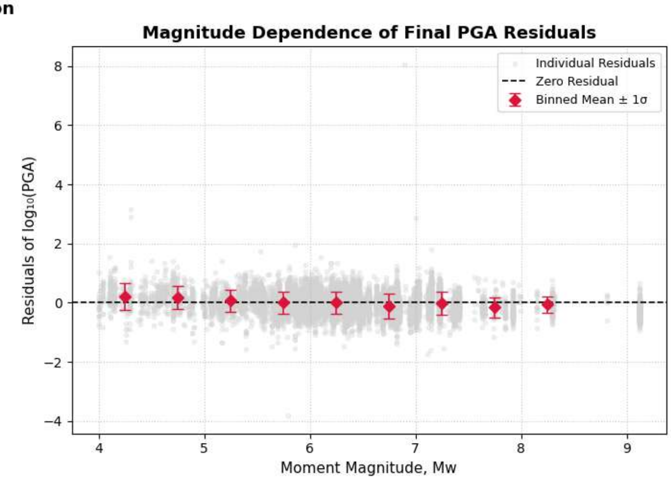


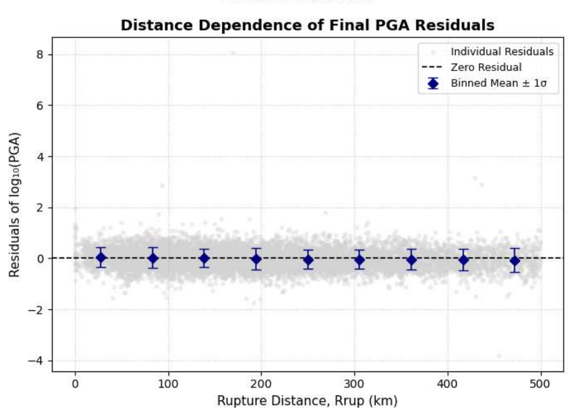
Final Test MSE: 0.1661
Final Test MAE: 0.3113
Final Overall R^2: 0.6660

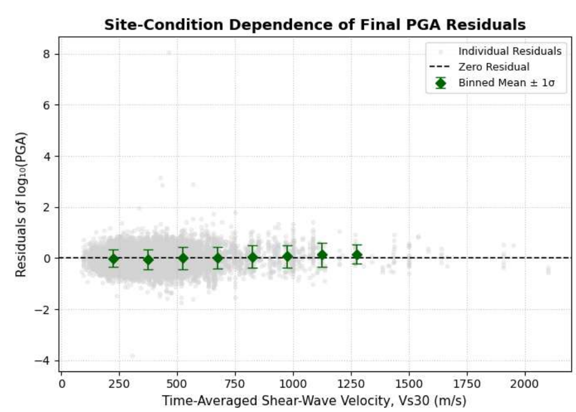
Final Per-output R^2 scores:
Output 00 (PGA_g): R^2 = 0.679
Output 01 (PGV_cm_sec): R^2 = 0.763

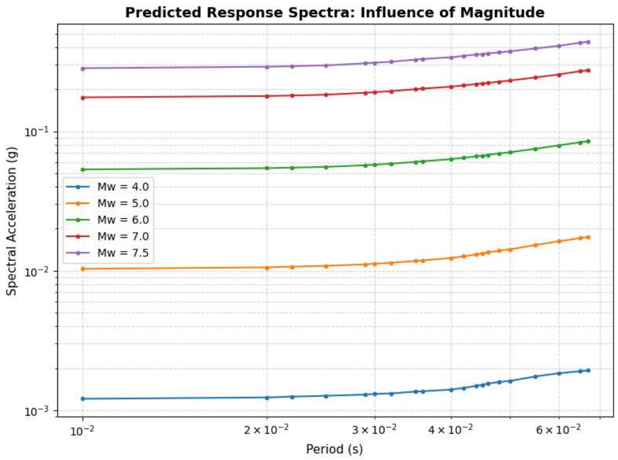
Output 02 (T0pt010S): R^2 = 0.678
Output 03 (T0pt020S): R^2 = 0.676
Output 04 (T0pt022S): R^2 = 0.676
Output 05 (T0pt025S): R^2 = 0.675
Output 06 (T0pt029S): R^2 = 0.673
Output 07 (T0pt030S): R^2 = 0.672
Output 08 (T0pt032S): R^2 = 0.671
Output 09 (T0pt035S): R^2 = 0.668
Output 10 (T0pt036S): R^2 = 0.667
Output 11 (T0pt040S): R^2 = 0.664
Output 12 (T0pt042S): R^2 = 0.662
Output 13 (T0pt044S): R^2 = 0.660
Output 14 (T0pt045S): R^2 = 0.659
Output 15 (T0pt046S): R^2 = 0.658
Output 16 (T0pt048S): R^2 = 0.656
Output 17 (T0pt050S): R^2 = 0.653
Output 18 (T0pt055S): R^2 = 0.647
Output 19 (T0pt060S): R^2 = 0.641
Output 20 (T0pt065S): R^2 = 0.635
Output 21 (T0pt067S): R^2 = 0.634

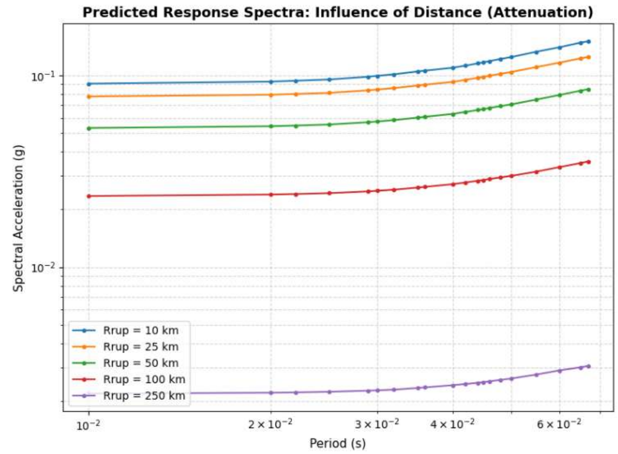


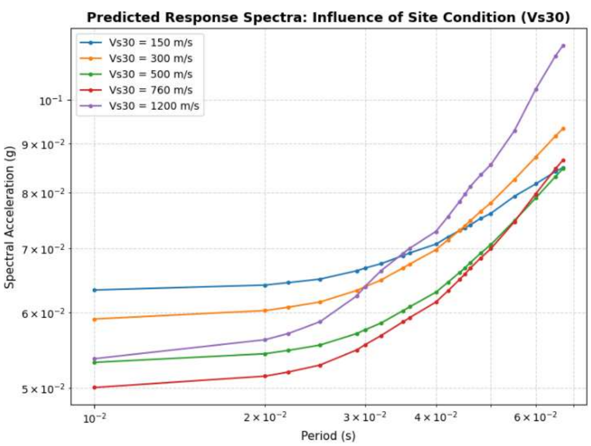


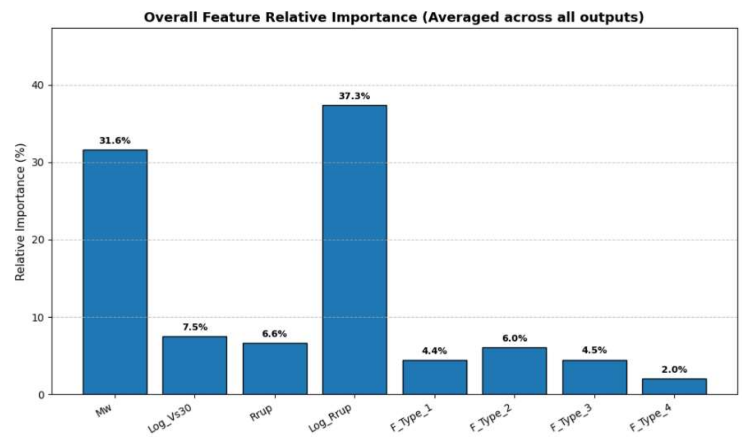


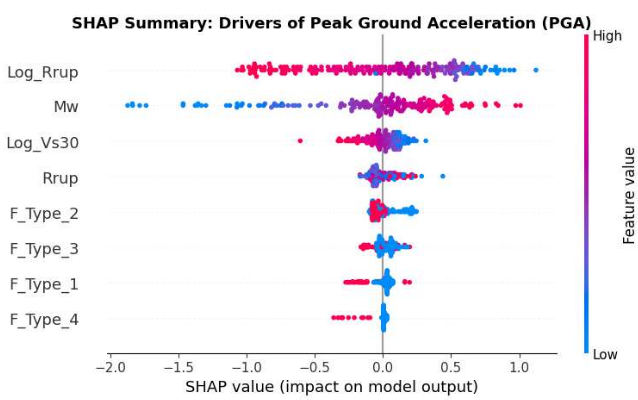


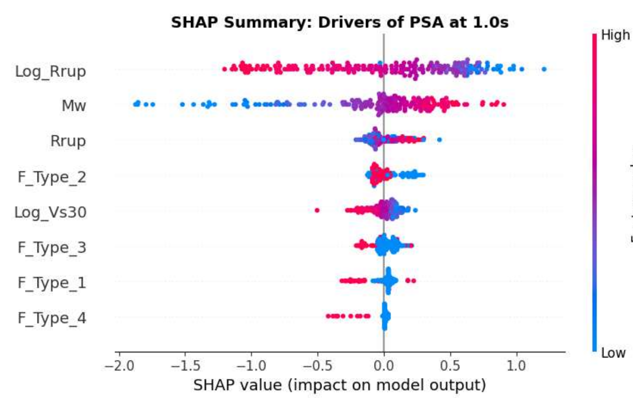












Inferences

1. Baseline Shallow NN

The baseline model demonstrates a strong initial capacity to capture the fundamental physics of seismic wave attenuation. With just a single hidden layer of 16 neurons, the network achieved a solid overall R^2, performing particularly well on velocity-driven parameters like PGV. However, the initial residual plots showed some dispersion. This indicates that while the base model understands the general trends, a minimal architecture of 16 neurons struggles to fully resolve the complex, non-